Molecular Graph Captioning
From Structural Encoders to LLM Fine-Tuning

ALTEGRAD 2025-26 Data Challenge

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The Challenge: Bridging Modalities

Objective

- **Task:** Generate natural language descriptions for molecular graphs.
- **Input:** Molecular Graph (Atoms, Bonds, Topology).
- **Output:** Textual description of chemical properties and structure.

Core Difficulty

- Bridging the gap between **invariant, non-Euclidean graphs** and **sequential, semantic text**.

Our Journey: 31 submissions iterating through Retrieval, Generative GNNs, and LLM Fine-Tuning.

Key Strategy Evolution

1. **Retrieval:** Latent space alignment.
2. **Generative GNN:** Custom Encoder-Decoder.
3. **LLM Transfer:** Fine-tuning BioT5.

Approach 1: Contrastive Retrieval (The Baseline)

Architecture

- **Graph Enc:** MoLFormer-XL.
- **Text Enc:** PubMedBERT.
- **Method:** InfoNCE Loss to maximize cosine similarity between matched pairs ($N = 128$).

Result: Score ≈ 0.60.

- Learned high-level classes but failed on specific substituents.
- Evaluated on R@1 and MRR
- Trained a re-ranker model but it didn't help

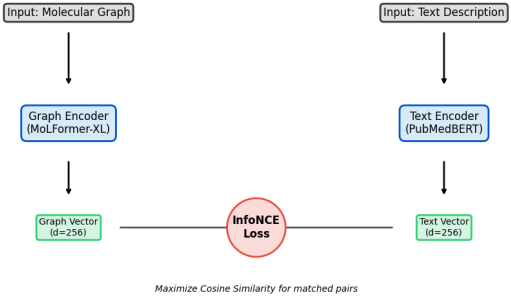
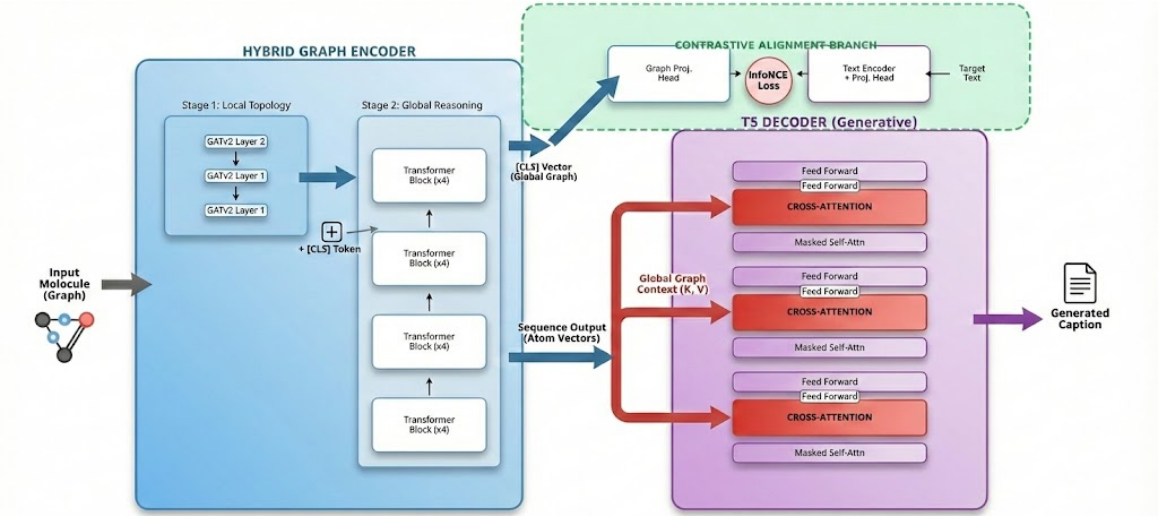


Figure: Retrieval Pipeline

Approach 2: Hybrid GNN-Transformer Encoder

The Architecture:



Approach 2 Deep Dive: The Hybrid Graph Encoder

We require an encoder that captures both specific chemical geometry and holistic molecular properties.

Stage 1: Local Topology (GATv2)

- **Input:** Atom features + Bond adjacency.
- **Mechanism:** 2 layers of Graph Attention Networks (GATv2).
- **Role:** Message passing between immediate neighbors. It "burns in" local chemical environment (e.g., aromaticity, functional groups) into node embeddings.

Stage 2: Global Reasoning (Transformer)

- **Input:** Node embeddings from GATv2 + a prepended learnable `[CLS]` token.
- **Mechanism:** 4-layer Transformer Encoder with full Self-Attention.
- **Role:** Allows every atom to attend to every other atom, capturing long-range dependencies regardless of graph distance.

Output: A sequence of refined atom vectors and one global `[CLS]` vector.

Approach 2 Deep Dive: The T5 Decoder & Cross-Attention

Why T5-Base outperforms Decoder-Only models (like Qwen)
A decoder-only model only sees the graph at the very start. T5, as an Encoder-Decoder, uses **Cross-Attention** to revisit the graph continuously.

The Cross-Attention Mechanism At decoding step t:

- **Query (Q):** The current text token being generated.
- **Key (K) & Value (V):** The *entire sequence* of atom embeddings from our Hybrid Graph Encoder.

The Result

The model dynamically focuses on relevant parts of the molecule (e.g., a specific ring structure) exactly when it needs to generate the corresponding word, preventing "catastrophic forgetting" of the input structure.

Approach 2 Deep Dive: Multi-Task Alignment Training

We train with two simultaneous objectives balanced by learnable uncertainty weights.

1. **Generative Loss ($\mathcal{L}_{gen}$):**
   - Cross-Entropy on T5 output tokens.
   - Teaches grammar and chemical syntax.
2. **Contrastive Loss ($\mathcal{L}_{cl}$):**
   - **Goal:** Align Global Graph (`[CLS]`) with Global Text.
   - **Method:** InfoNCE loss maximizing cosine similarity.

Loss Function

We used **Learnable Uncertainty Weights** to balance objectives:

$$\mathcal{L}_{total} = \frac{1}{2\sigma_g^2}\mathcal{L}_{gen} + \frac{1}{2\sigma_c^2}\mathcal{L}_{cl} + \log(\sigma_g \sigma_c)$$

Approach 3: BioT5 Fine-Tuning (The Winner)

Rationale

- BioT5 is pre-trained on massive biological corpora and PubMed abstracts.
- Strong priors on chemical syntax vs. learning *tabula rasa*.

Implementation

- **Input:** SELFIES (Self-Referencing Embedded Strings).
- **Benefit:** 100% valid molecular graphs; robust tokenization.
- **Training:** 30 epochs, BFloat16.

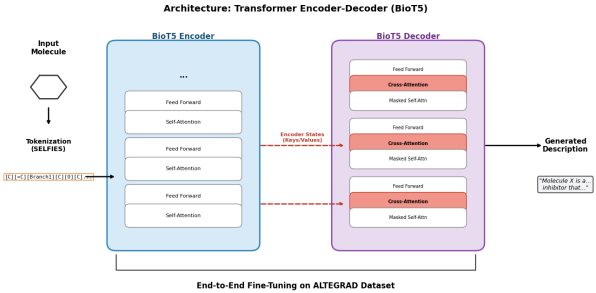


Figure: BioT5 finetuning

Quantitative Results Summary

Method	Key Feature	Decoder	Score
Retrieval Baseline	Contrastive (MoLFormer)	N/A	0.60
Generative GNN (v1)	Mean Pooling	T5-Base	0.62
Generative GNN (v2)	**Global Attention (CLS)**	**T5-Base**	0.64
BioT5 Fine-Tuning	**SELFIES Tokenization**	**BioT5**	**0.79**

Table: Progression of submissions. The semantic knowledge transfer of BioT5 yielded the largest performance jump.

Discussion: Key Takeaways

1. **Architecture Matters:**
   - **Encoder-Decoder > Decoder-Only:** Cross-Attention is essential for maintaining fidelity to the source graph throughout generation.
   - **Attention > Pooling:** Global attention (`[CLS]`) captures rare functional groups better than mean pooling.

2. **The Power of Pre-training:**
   - Structural encoders (GNNs) offer strong topological bias.
   - However, they cannot compete with the semantic richness of domain-specific LLMs (BioT5) for captioning tasks.
   - **Conclusion:** Treat captioning as a translation task (SELFIES $\rightarrow$ Text).

Conclusion and Future Work

Achievements

- Conducted extensive search (31 submissions).
- Successfully diagnosed "Attention Decay" in causal models.
- Achieved **0.79** on the leaderboard via BioT5.

Future Directions

- **Graph-Injected LLMs:** Fusing GNN embeddings directly into BioT5 cross-attention layers.
- Combining the topological precision of Approach 2 with the semantic fluency of Approach 3.

Thank you for your attention!